

Artificial Intelligence Search

Lecture: 5
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Portions of this lecture are adapted from the course materials of Prof. Sudeshna Sarkar, Department of Computer Science and Engineering, IIT Kharagpur.

Problem solving as search

- Goal formulation
 - Problem formulation
 - Actions
 - States
-
- Before actually doing something to solve a puzzle, an intelligent agent explores possibilities “in its head”
 - Search = “mental exploration of possibilities”
 - Making a good decision requires exploring several possibilities
 - Execute the solution once it's found

Formulating Problems as Search

*Given an **initial state** and a **goal**, find the sequence of **actions** leading through a sequence of **states** to the final **goal state**.*

Terms:

- **Successor function**: given action and state, returns {**action**, **successors**}
- **State space**: the set of all states reachable from the initial state
- **Path**: a sequence of states connected by actions
- **Goal test**: is a given state the goal state?
- **Path cost**: function assigning a numeric cost to each path
- **Solution**: a path from initial state to goal state

Search Problems

- A search problem consists of:
 - A state space
 - A successor function / Operators [and costs]
 - Start state
 - Goal state [test]
- **Output: Path: start $\rightarrow$ a state satisfying goal test**
- A solution is a sequence of actions (a plan) which transforms the start state to a goal state

Representing states

- What information is necessary to encode about the world to sufficiently describe all relevant aspects to solving the goal? That is, what knowledge needs to be represented in a state description to adequately describe the current state or situation of the world?
- The **size of a problem** is usually described in terms of the **number of states** that are possible.
 - Tic-Tac-Toe has about 3^9 states.
 - Checkers has about 10^{40} states.
 - Rubik's Cube has about 10^{19} states.
 - Chess has about 10^{120} states in a typical game.

Some example problems

- Toy problems and micro-worlds

- 8-Puzzle
- Missionaries and Cannibals
- Cryptarithmic
- Remove 5 Sticks
- Water Jug Problem

- Real-world problems

8-Puzzle

Given an initial configuration of 8 numbered tiles on a 3 x 3 board, move the tiles in such a way so as to produce a desired goal configuration of the tiles.

5	4	
6	1	8
7	3	2

Start State

1	2	3
8		4
7	6	5

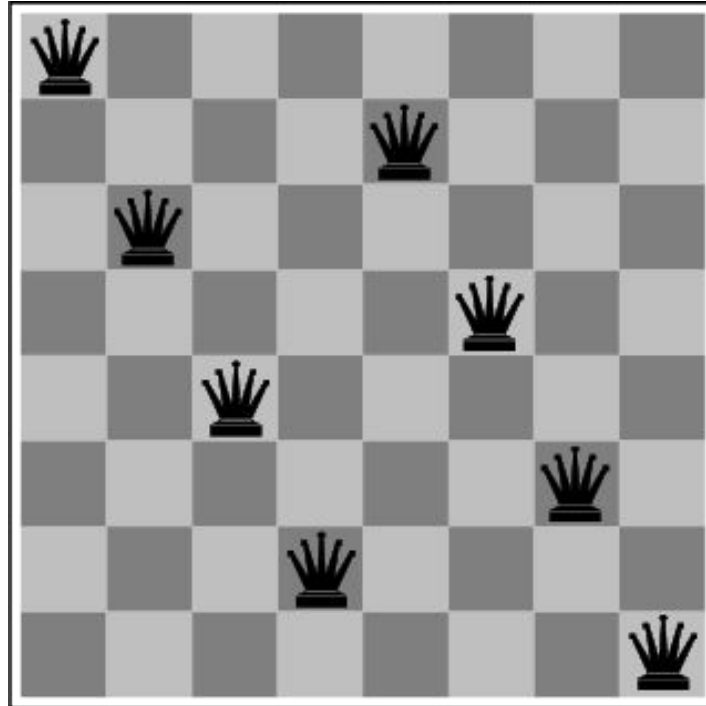
Goal State

8 Puzzle

- **States:** integer locations of tiles
 - (0 1 2 3 4 5 6 7 B)
 - (0 1 2)(3 4 5) (6 7 B)
- **Action:** left, right, up, down
- **Goal test:** is current state = (0 1 2 3 4 5 6 7 B)?
- **Path cost:** same for all paths
- **Successor function:** given {up, (5 2 3 B 1 8 4 7 6)} -> ?
- What would the **state space** be for this problem?

The 8-Queens Problem

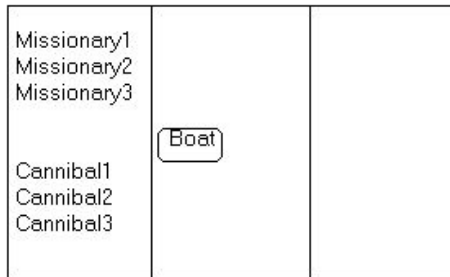
Place eight queens on a chessboard such that no queen attacks any other!



Missionaries and Cannibals

There are 3 missionaries, 3 cannibals, and 1 boat that can carry up to two people on one side of a river.

- **Goal:** Move all the missionaries and cannibals across the river.
- **Constraint:** Missionaries can never be outnumbered by cannibals on either side of river, or else the missionaries are killed.
- **State:** configuration of missionaries and cannibals and boat on each side of river.
- **Operators:** Move boat containing some set of occupants across the river (in either direction) to the other side.



3 Missionaries and 3 Cannibals wish to cross the river. They have a boat that will carry two people. Everyone can navigate the boat. If at any time the Cannibals outnumber the missionaries on either bank of the river, they will eat the Missionaries. Find the smallest number of crossings that will allow everyone to cross the river safely.

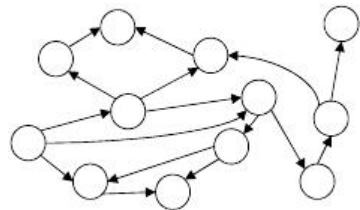
The problem can be solved in 11 moves. But people rarely get the optimal solution, because the MC problem contains a 'tricky' state at the end, where two people move back across the river.

Missionaries and Cannibals Solution

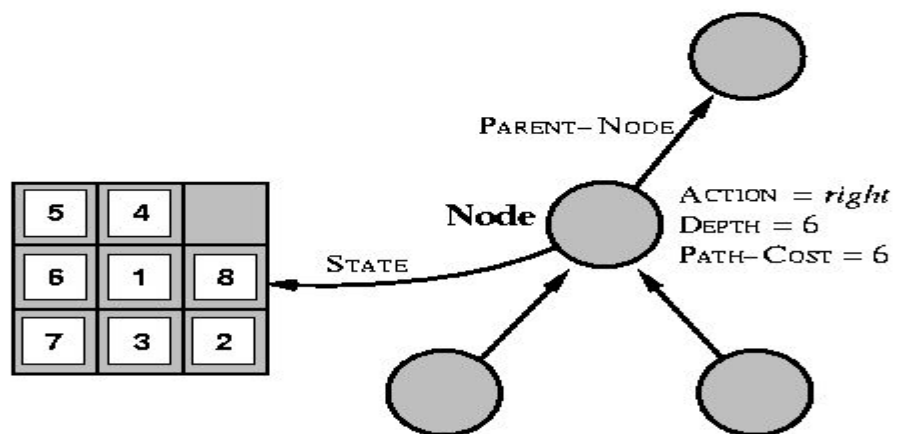
	<u>Near side</u>	<u>Far side</u>
0 Initial setup:	MMMCCC B	-
1 Two cannibals cross over:	MMMC	B CC
2 One comes back:	MMMCC B	C
3 Two cannibals go over again:	MMM	B CCC
4 One comes back:	MMMC B	CC
5 Two missionaries cross:	MC	B MMCC
6 A missionary & cannibal return:	MMCC B	MC
7 Two missionaries cross again:	CC	B MMMC
8 A cannibal returns:	CCC B	MMM
9 Two cannibals cross:	C	B MMMCC
10 One returns:	CC B	MMMC
11 And brings over the third:	-	B MMMCCC

Formalizing search in a state space

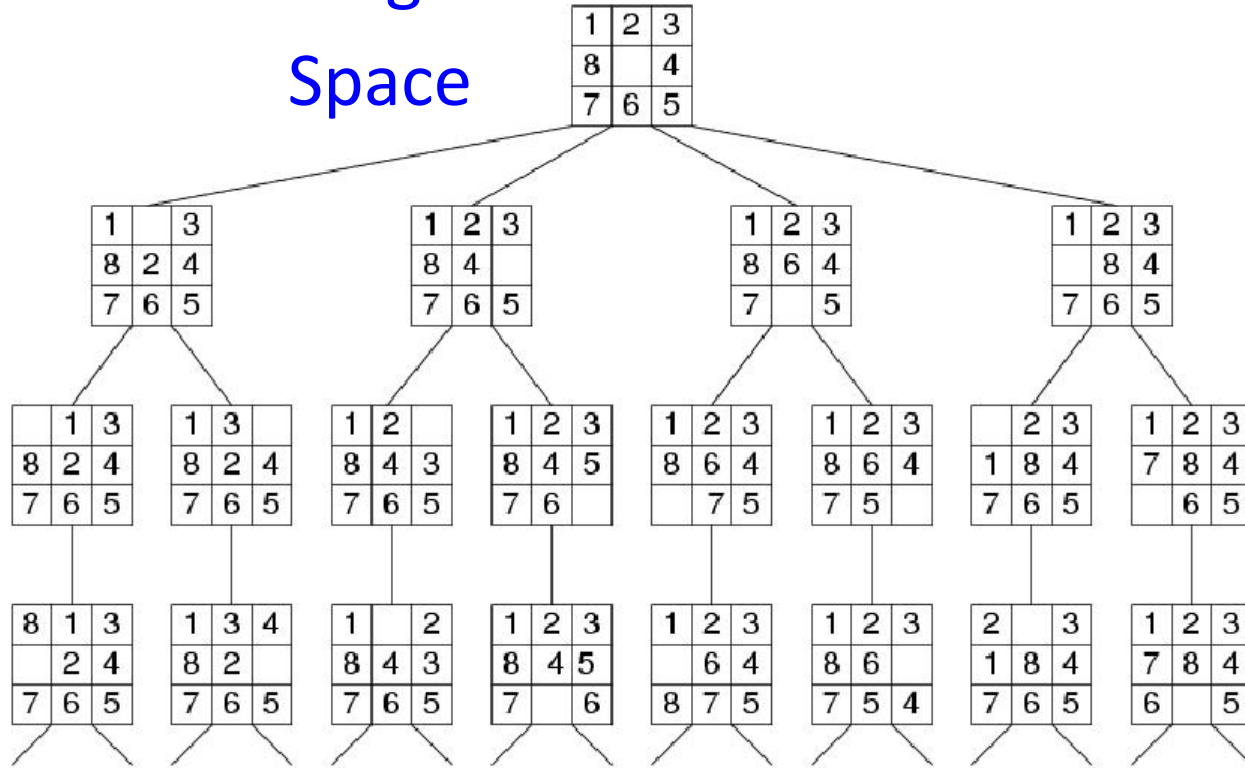
- A state space is a **graph** (V, E)
- Each **node** is a data structure
 - a state description
 - the parent of the node,
 - the operator that generated the node from that parent
 - ...depth, path cost,



- Expand
 - A function that given a node, creates all children nodes, using **successor function**



Search Tree Example: Fragment of 8-Puzzle Problem Space



Search strategies

- A search strategy is defined by picking the **order of node expansion**. Strategies are evaluated along the following dimensions:
- **Completeness:** Guarantees finding a solution whenever one exists
- **Time complexity:**
- **Space complexity:**
- **Optimality/Admissibility**
- Time and space complexity are measured in terms of
 - b : maximum branching factor of the search tree
 - d : depth of the least-cost solution
 - m : maximum depth of the state space (may be ∞)